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07 — DCE-MRI of the Kidney: Computational Imaging and Modeling for GFR Estimation

BMED365 – Computational Medicine (Lab 5)

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Based on: 01-dce-mri-kidney-explore.ipynb (BMED360-2021) and functional-kidney-imaging (ESMRMB 2016)
Updated: 2026-02-19 (Cursor 2.5.17 with Claude Opus 4.6)
Updated: 2026-02-20 (Cursor 2.5.20 with Claude Opus 4.6) - saved notebook to .pdf

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> **Prerequisites**: You should have worked through 06-kidney-filtration.ipynb, which introduces the physiology of glomerular filtration and the concept of GFR. This notebook extends that foundation into the realm of **medical imaging** — specifically **Dynamic Contrast-Enhanced MRI (DCE-MRI)** — to explore how GFR can be estimated non-invasively from image data.

1. Motivation & Learning Objectives

Why image-based GFR estimation?

In notebook 06, we saw that **Glomerular Filtration Rate (GFR)** is the single most important measure of kidney function. Traditionally, GFR is measured using blood and urine tests (inulin clearance, creatinine clearance) or estimated from serum creatinine using equations (CKD-EPI). However, these methods provide only a **global** (whole-body) estimate — they cannot tell us:

- **Which kidney** is functioning better (critical for surgical planning)
- **Where** within a kidney the function is impaired (focal vs. diffuse disease)
- **How** kidney function changes dynamically over seconds to minutes

Dynamic Contrast-Enhanced MRI (DCE-MRI) addresses all these limitations by:

1. **Imaging both kidneys simultaneously** with high spatial resolution
2. **Tracking** the passage of a gadolinium-based contrast agent (GBCA) through the renal vasculature and tubules over time
3. **Applying tracer kinetic models** (compartment models) to the time-varying signal to estimate physiological parameters — including **single-kidney GFR** (skGFR)

This makes DCE-MRI a powerful tool in computational medicine: it combines **physics** (MRI signal formation), **physiology** (renal hemodynamics), **mathematics** (tracer kinetics, ODEs), and **computation** (curve fitting, motion correction, clustering) into a single quantitative pipeline.

Clinical need

- **Chronic Kidney Disease (CKD)** affects >800 million people globally (~10–15% of adults)
- **Living kidney donation** requires accurate assessment of individual kidney function
- **Renal artery stenosis**, **hydronephrosis**, and **renal tumors** can cause asymmetric kidney damage
- **Pediatric nephrology** needs radiation-free methods (unlike nuclear medicine)
- Standard eGFR equations have limited accuracy in extremes of body composition, age, and diet

What you will learn

In this notebook you will:

1. **Understand** what DCE-MRI is and how it measures kidney function
2. **Explore** real 4D (3D + time) DCE-MRI data of the kidney
3. **Appreciate** the challenges of **motion correction** in abdominal DCE-MRI
4. **Apply K-means clustering** to segment kidney tissue based on signal intensity time courses ($\mathbf{R}^{50}$ feature space)
5. **Use K-means "smoothing"** to create prototypic time courses that reduce noise
6. **Implement tracer kinetic models** (compartment models) to estimate renal perfusion and GFR
7. **Understand** the historical development and clinical translation of DCE-MRI-based GFR estimation

2. What is DCE-MRI? — Definitions and Key Concepts

2.1 Dynamic Contrast-Enhanced MRI (DCE-MRI)

DCE-MRI is an MRI technique where:

1. A **gadolinium-based contrast agent (GBCA)** is injected intravenously as a bolus
2. Rapid, repeated MRI acquisitions (typically every 2–5 seconds) capture the **time-varying signal** as the contrast agent passes through the tissue
3. The resulting **4D dataset** (3D spatial + time) encodes information about tissue perfusion, permeability, and filtration

Term	Definition
GBCA	Gadolinium-Based Contrast Agent — a paramagnetic substance that shortens T_1 relaxation time, causing signal enhancement on T_1-weighted images
Bolus	A single, rapid injection of contrast agent (typically 0.05–0.1 mmol/kg body weight)

Term	Definition
Temporal resolution	Time between successive 3D volume acquisitions (Δt ≈ 2–5 s for renal DCE-MRI)
Signal intensity time course	S(t) — the MRI signal measured in a voxel or region as a function of time
Concentration time course	C(t) — the contrast agent concentration derived from S(t) using a signal model
Arterial Input Function (AIF)	$C_a(t)$ — the concentration time course in a feeding artery (e.g., aorta or renal artery)

2.2 From Signal to Concentration

The MRI signal in a T_1-weighted spoiled gradient echo (SPGR) sequence depends on the longitudinal relaxation rate $R_1 = 1/T_1$:

$$S(t) = S_0 \cdot \frac{\sin\alpha \cdot (1 - e^{-TR \cdot R_1(t)})}{1 - \cos\alpha \cdot e^{-TR \cdot R_1(t)}}$$

where α is the flip angle and TR is the repetition time. The contrast agent increases R_1 linearly:

$$R_1(t) = R_{1,0} + r_1 \cdot C(t)$$

where $R_{1,0}$ is the baseline (pre-contrast) relaxation rate and r_1 is the **relaxivity** of the contrast agent (≈ 3.7–4.5 $L{\cdot}mmol^{-1}{\cdot}s^{-1}$ at 1.5 T for typical GBCAs).

2.3 Tracer Kinetics and Compartment Models

A **compartment model** represents the tissue as a set of well-mixed compartments connected by rate constants. For the kidney, the key compartments are:

Compartment	Physical meaning	Symbol
Vascular (plasma)	Blood plasma in glomerular and peritubular capillaries	v_p (fractional volume)
Tubular	Fluid in proximal tubules (after filtration)	v_t (fractional volume)
Interstitial	Extracellular, extravascular space	v_e (fractional volume)

The **two-compartment filtration model** for the kidney (Sourbron et al., 2008; Annet et al., 2004) describes contrast agent transport as:

$$C_{tissue}(t) = v_p \cdot C_a(t) + GFR \cdot \int_0^t C_a(\tau)\, d\tau - k_{out} \cdot \int_0^t C_{tissue}(\tau)\, d\tau$$

where:

- v_p = plasma volume fraction
- GFR = glomerular filtration rate (per unit tissue volume, mL/min/mL)

- k_{out} = tubular outflow rate constant (related to transit time through the tubules)

2.4 Glomerular Filtration Rate (GFR) from DCE-MRI

The **single-kidney GFR** (skGFR) is obtained by:

$$\text{skGFR} = GFR_{density} \times V_{kidney}$$

where $GFR_{density}$ is the filtration rate per unit volume of functioning parenchyma and V_{kidney} is the kidney volume (measured from the MRI images).

Reference values:

- Total GFR (both kidneys): ~90–120 mL/min (age- and sex-dependent)
- Single-kidney GFR: ~45–60 mL/min (assuming symmetric function)

2.5 The Measurement Pipeline

The complete DCE-MRI pipeline for kidney GFR estimation involves:

```
MRI Acquisition  →  Motion Correction  →  Signal-to-Concentration  →
AIF Extraction

↓
    GFR Map  ←  Compartment Model Fitting  ←  ROI / Voxel-wise
Analysis  ←
```

Each step introduces potential errors and requires careful computational methods.

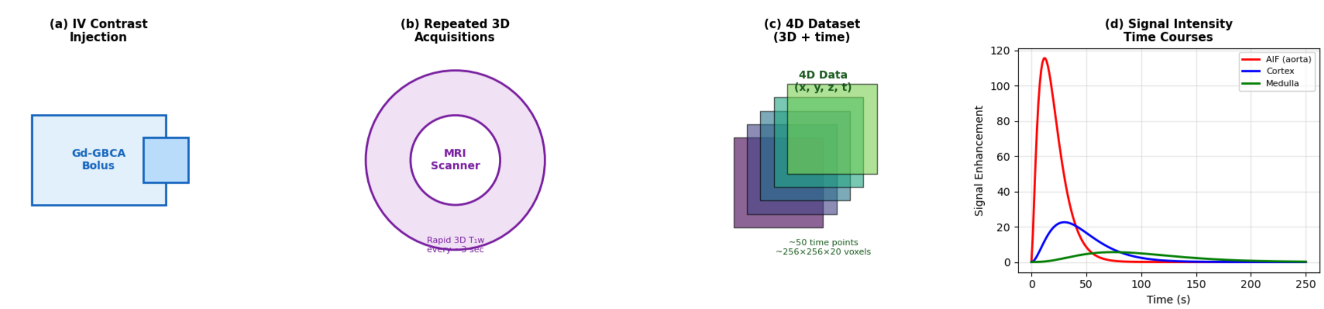


Figure 1. Schematic overview of DCE-MRI kidney acquisition

3. Historical Context

The road to DCE-MRI-based GFR estimation

The idea of using dynamic imaging with exogenous tracers to quantify organ function has a rich history spanning nuclear medicine, CT, and MRI.

Year	Milestone	Key Contributors
1927	**Indicator dilution theory** — mathematical framework for tracer kinetics	Hamilton, Moore, Kinsman
1954	**Renography** — first dynamic nuclear imaging of the kidneys using ^{131}I-hippuran	Taplin, Meredith, Kade, Winter
1960s	**Compartment modeling** formalized for pharmacokinetics	Teorell, Rescigno, Segre
1984	**First DCE-MRI** studies using Gd-DTPA in brain tumors	Weinmann, Brasch, Runge
1991	**Tofts model** — standard pharmacokinetic model for DCE-MRI	Paul **Tofts** & Allan **Kermode**
1994	First DCE-MRI of the kidneys with Gd-DTPA for renal function assessment	Prigent, Cosgriff, Gates
1999	**Extended Tofts model** including vascular term v_p	Paul **Tofts** et al.
2004	**Annet model** — two-compartment (vascular + tubular) model for renal DCE-MRI	Laurent **Annet** et al.
2008	**Sourbron separable compartment model** for renal perfusion and filtration	Steven **Sourbron** et al.
2012	ISMRM Perfusion Study Group consensus on DCE-MRI nomenclature	Tofts, Sourbron, Buckley et al.
2014	**Hodneland et al.** — segmentation-driven image registration for kidney DCE-MRI motion correction	Erlend **Hodneland** et al.
2015	**Eikefjord et al.** — comprehensive framework for DCE-MRI GFR estimation at 1.5T	Eli **Eikefjord** et al.
2018	PARENCHIMA COST Action — European consortium for standardization of renal MRI	Nils **Grenier**, Steven **Sourbron** et al.
2020	Consensus paper on technical recommendations for clinical renal MRI	Mendichovszky, Notohamiprodjo et al.
2023–26	AI/ML approaches for automated DCE-MRI analysis, AIF detection, motion correction	Multiple groups

Paul Tofts and the pharmacokinetic model

Paul S. Tofts (b. 1947) at University College London developed the most widely used pharmacokinetic model for DCE-MRI analysis. His 1991 paper with Kermode established the framework for quantitative analysis of blood-brain barrier permeability from DCE-MRI. The **Tofts model** and its extensions remain the foundation of DCE-MRI analysis across organs.

From brain to kidney: the challenges

While DCE-MRI was first developed for brain tumors, applying it to the **kidneys** introduced unique challenges:

1. **Respiratory motion** — the kidneys move 1–3 cm with each breath, requiring motion correction or breath-hold protocols

2. **Complex anatomy** — cortex and medulla have very different enhancement patterns
3. **Filtration** — unlike most tissues where contrast leaks passively, the kidney actively **filters** contrast through the glomeruli
4. **Rapid dynamics** — renal enhancement peaks within 15–30 seconds, requiring high temporal resolution

Bergen contributions

The University of Bergen group (Lundervold, Eikefjord, Hodneland, Rørvik, Svarstad) has made significant contributions to renal DCE-MRI:

- **Segmentation-driven motion correction** (Hodneland et al., IEEE TIP 2014)
- **Comprehensive GFR estimation framework** comparing FLASH and KWIC acquisitions (Eikefjord et al., AJR 2015)
- **Repeatability, accuracy, and precision** of DCE-MRI GFR using iohexol-GFR as reference (Eikefjord et al., Acta Radiol 2017)

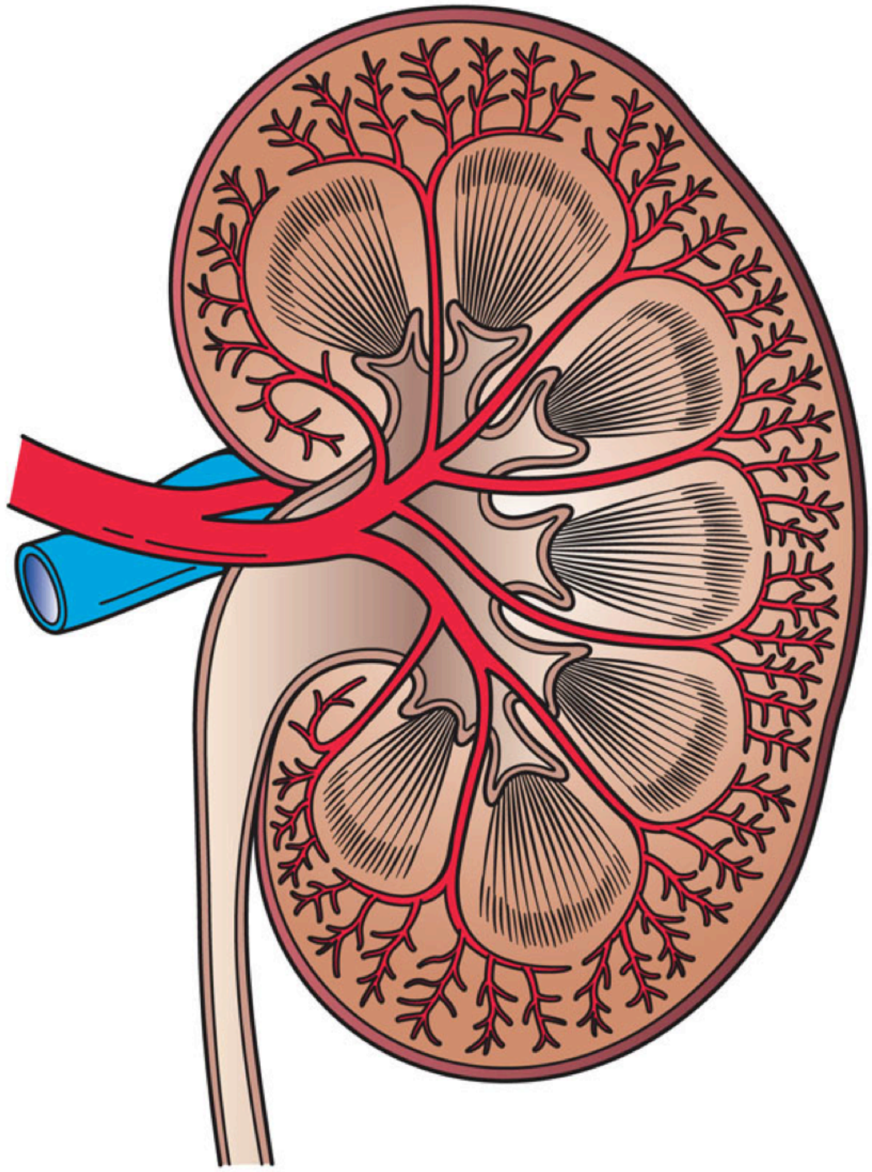


Figure 2. Cross-section of the human kidney. In DCE-MRI, the contrast agent enters via the **renal artery**, passes through the cortical **glomeruli** (where it is filtered), and transits through the **medullary** tubules before being excreted via the **collecting system**. The cortex enhances rapidly (high perfusion), while the medulla enhances more slowly (tubular transit).
Source: Wikimedia Commons (Holly Fischer / Piotr Michał Jaworski, CC BY 3.0 / CC BY-SA 3.0).

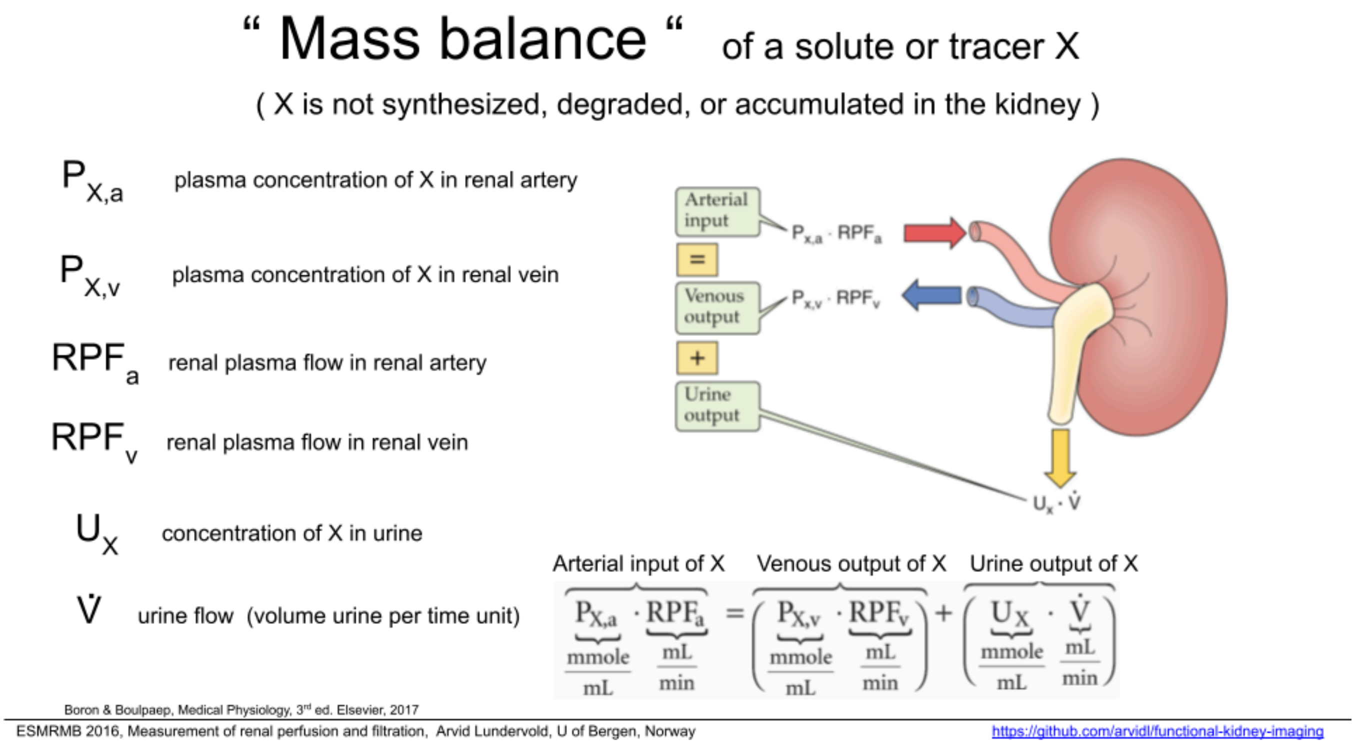


Figure. Mass balance and kidney filtration in the context of DCE-MRI. The principle of *conservation of mass* (indicator dilution theory) underlies the tracer kinetic models used to estimate GFR from DCE-MRI. The contrast agent concentration in tissue reflects the balance between arterial inflow, glomerular filtration into the tubules, and venous/urinary outflow.
Source: BMED360 Lecture 6 — Vascular permeability and DCE-MRI (A. Lundervold, UiB).

4. Scientific and Clinical Implications

Why DCE-MRI matters for kidney disease

Clinical scenario	Limitation of eGFR	DCE-MRI advantage
Kidney donor evaluation	Cannot assess individual kidney function	Provides **single-kidney GFR**
Renal artery stenosis	May be falsely normal if one kidney compensates	Shows **asymmetric** function loss
Partial nephrectomy planning	Cannot predict remaining function after surgery	Maps **regional** function
Pediatric CKD	eGFR equations less reliable in children	**Radiation-free**, accurate, repeatable
Transplant monitoring	Slow creatinine response to acute rejection	**Real-time** perfusion changes
Drug nephrotoxicity	Serum creatinine rises only after significant damage	May detect **early** functional changes

The computational medicine perspective

DCE-MRI-based GFR estimation is a prime example of **computational medicine** in action:

1. **Physics** → MRI signal formation (Bloch equations, relaxation, contrast agent relaxivity)
2. **Engineering** → Fast pulse sequences (FLASH, KWIC, radial), parallel imaging, motion correction
3. **Mathematics** → Tracer kinetics (convolution integrals, ODEs), compartment models, curve fitting
4. **Computer science** → Image registration, K-means clustering, optimization algorithms, machine learning
5. **Physiology** → Renal hemodynamics, glomerular filtration, tubular function
6. **Clinical medicine** → Diagnosis, treatment planning, monitoring

This notebook sits at the intersection of all these disciplines — exactly what computational medicine is about.

Check Your Understanding: Motivation and Concepts

Q1. What key limitation of serum creatinine-based eGFR does DCE-MRI overcome?

▶ Click to reveal answer

Q2. Name the three main tissue compartments in a kidney DCE-MRI compartment model and explain what each represents.

▶ Click to reveal answer

Q3. Why is motion correction particularly important in kidney DCE-MRI compared to brain DCE-MRI?

▶ Click to reveal answer

5. Setup and Data

Required packages

This notebook uses standard scientific Python packages plus `nibabel` for neuroimaging file I/O, `scikit-learn` for K-means clustering, and `gdown` for downloading data from Google Drive.

The DCE-MRI dataset

We use a **4D NIfTI** file from a clinical DCE-MRI kidney examination:

- **Patient**: Anonymized subject from a renal function study
- **Scanner**: 1.5T MRI (Siemens)
- **Sequence**: 3D T_1-weighted spoiled gradient echo (FLASH)
- **Contrast agent**: Gd-DTPA (Magnevist), 0.05 mmol/kg
- **Temporal resolution**: ~3 seconds per 3D volume
- **Dimensions**: 256 × 256 × 20 slices × ~50 time points (= ~65 million voxels total)
- **File**: `kidney_capio_20050419_series_13_32.nii.gz`

```
All imports successful
  Platform: Darwin (arm64)
  NumPy:    2.2.6
  nibabel:  5.3.2

./data exists already!
./assets exists already!

Data file: /Users/arvid/GitHub/BMED365-2026/Lab5-Comp-Mod/notebooks/data/kidney_capio
_20050419_series_13_32.nii.gz
File exists: True

============================================================
4D DCE-MRI Dataset Summary
============================================================
  Shape:       (256, 256, 20, 20)
  Dimensions:  256x256 in-plane, 20 slices, 20 time points
  Voxel size:  (np.float32(1.484375), np.float32(1.484375), np.float32(3.0)) mm
  TR (approx): 2.5 s
  Data type:   float64
  Value range: [0.0, 143.0]
  Memory:      209.7 MB

  Total voxels per time point: 1,310,720
  Total voxel-time measurements: 26,214,400
```

6. Exploring the 4D DCE-MRI Data

Let's visualize the DCE-MRI data to understand its structure. We'll look at:

1. **Anatomical view** — a single time point to see the kidney anatomy
2. **Temporal montage** — the same slice at different time points to see contrast dynamics
3. **Signal intensity time courses** — S(t) curves from different tissue types

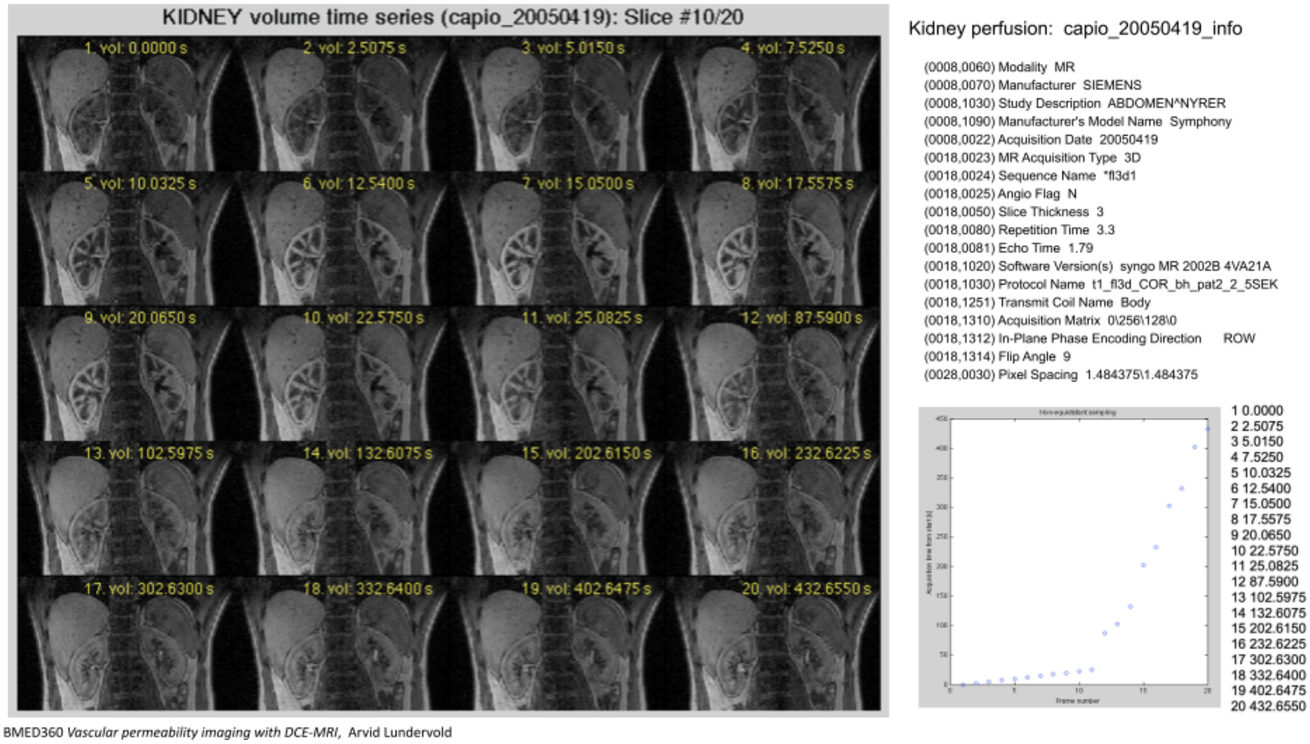


Figure. 3D DCE-MRI kidney volume time series from a clinical examination (Capio, 2005). Each panel shows the same coronal slice at successive time points after Gd-DTPA injection. Note the progressive enhancement of the renal cortex (bright rim), followed by medullary enhancement, and finally excretion into the collecting system (bright pelvis). The respiratory motion between frames is clearly visible — motivating the need for motion correction before quantitative analysis.

Source: BMED360 Lecture 6 — Vascular permeability and DCE-MRI (A. Lundervold, UiB).

Figure 3. Axial slice 10 at different DCE-MRI phases

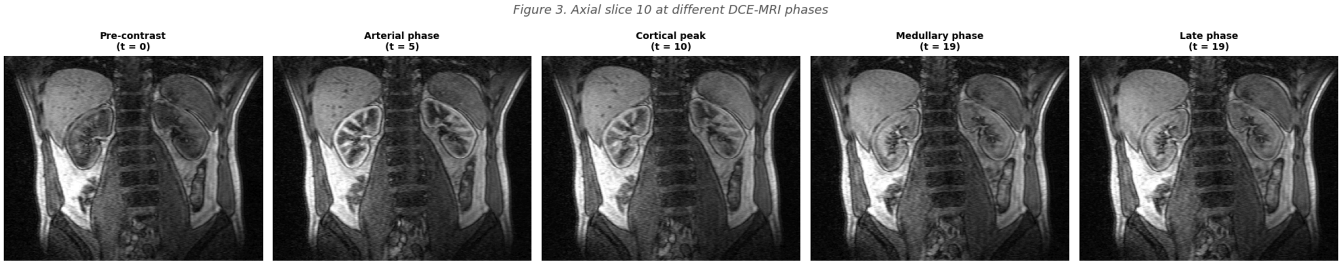


Figure 4. Temporal montage: slice 10, all 20 time points
Watch the contrast agent arrive (bright enhancement) and wash out

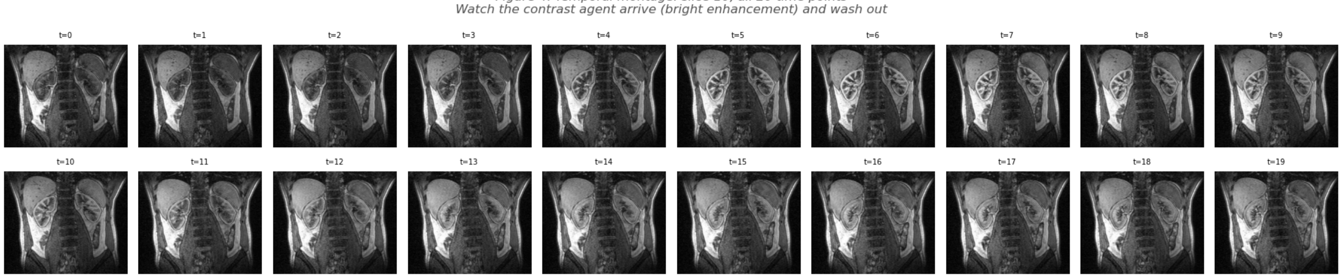
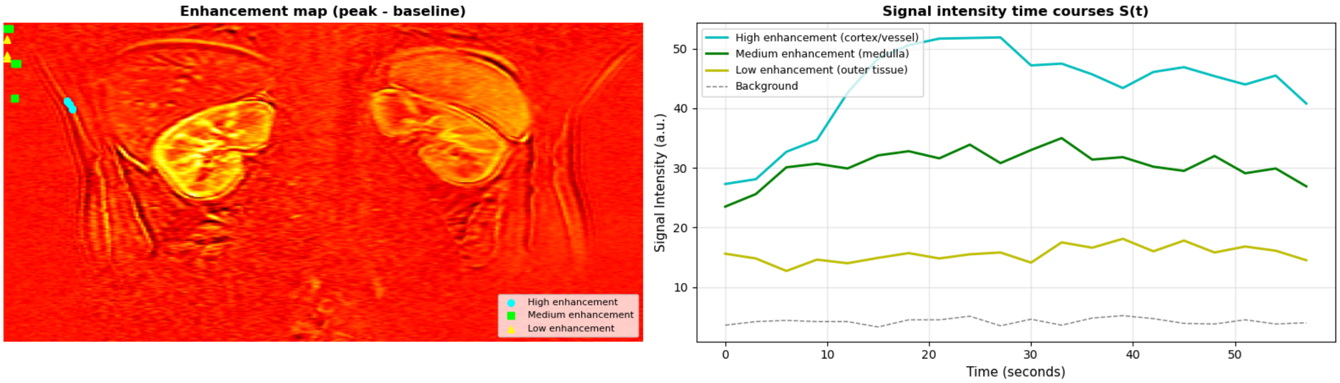


Figure 5. ROI-based signal intensity time courses from a DCE-MRI slice



7. Motion Correction in Kidney DCE-MRI

The problem

During a DCE-MRI acquisition lasting 3–5 minutes, the patient breathes ~40–60 times. Each breath cycle causes the kidneys to translate **1–3 cm** in the craniocaudal (head-to-foot) direction, and also to deform slightly due to compression by the diaphragm and liver.

Without motion correction, a single voxel's time course may sample:

- Renal cortex at time t_1
- Renal medulla at time t_2
- Perirenal fat at time t_3

This makes the time course **physiologically meaningless** for compartment model fitting.

Motion correction strategies

Strategy	Description	Pros	Cons
Breath-hold	Patient holds breath during acquisition	Simple, no post-processing	Limited to ~20 s; poor temporal coverage
Respiratory gating	Acquire only during one phase of respiration	Good image quality	Long scan times; irregular sampling
Rigid registration	Translate/rotate each 3D volume to align with a reference	Fast; handles bulk motion	Does not handle deformation
Deformable registration	Non-rigid warping of each volume	Handles organ deformation	Computationally expensive; risk of artifacts
Segmentation-driven	Use organ segmentation to guide registration	Robust; physiologically informed	Requires good segmentation

The **segmentation-driven deformable registration** approach by Hodneland et al. (2014) combines organ segmentation with non-rigid registration, using the kidney boundary to constrain the deformation field. This approach was developed specifically for kidney DCE-MRI at the University of Bergen.

Illustration: what motion does to time courses

Figure 6. Motion correction in kidney DCE-MRI

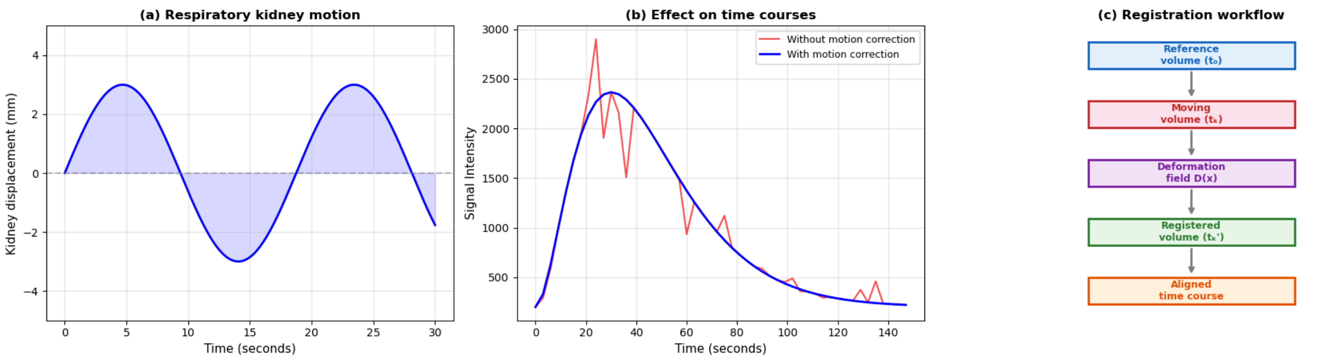
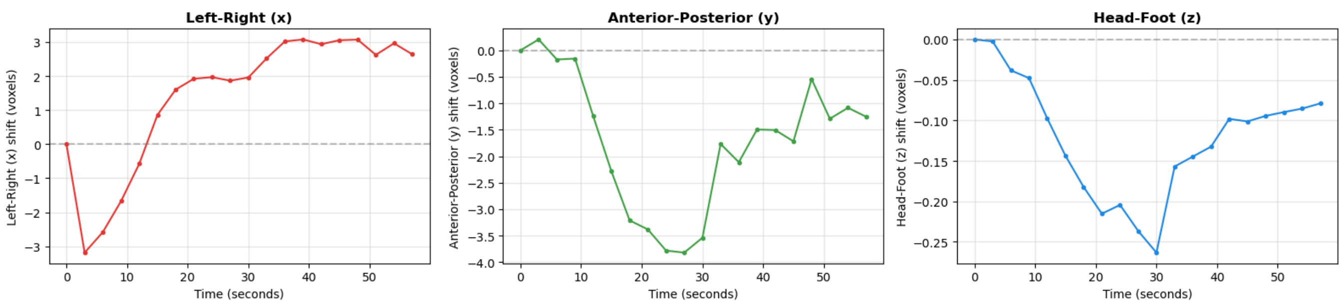


Figure 7. Center-of-mass displacement over time (proxy for respiratory motion)




```
Maximum displacement (voxels): x=6.3, y=4.0, z=0.3
```

DIY: Motion Correction Concepts

Q4. If a kidney voxel has size 1.5 × 1.5 × 3.0 mm and respiratory motion causes 15 mm of craniocaudal displacement, how many voxels does the kidney move in the head-foot direction?

▶ Click to reveal answer

Q5. What is the key difference between **rigid** and **deformable** registration? When would you prefer one over the other for kidney DCE-MRI?

▶ Click to reveal answer

8. K-means Clustering of Signal Intensity Time Courses

Motivation: Using time courses as feature vectors

Each voxel in the 4D DCE-MRI dataset has a **signal intensity time course** — a vector in $\mathbf{R}^{N_t}$ (e.g., $\mathbf{R}^{50}$ if there are 50 time points). The idea is elegant:

> **Voxels with similar physiology produce similar time courses.**

- Cortical voxels (high perfusion, rapid enhancement and washout) cluster together
- Medullary voxels (delayed enhancement, slow washout) cluster together
- Vascular voxels (very rapid, sharp peak) cluster together
- Background voxels (no enhancement) cluster together

K-means clustering groups voxels based on the **similarity of their entire time course**, treating each time course as a point in high-dimensional feature space ($\mathbf{R}^{N_t}$).

The K-means algorithm

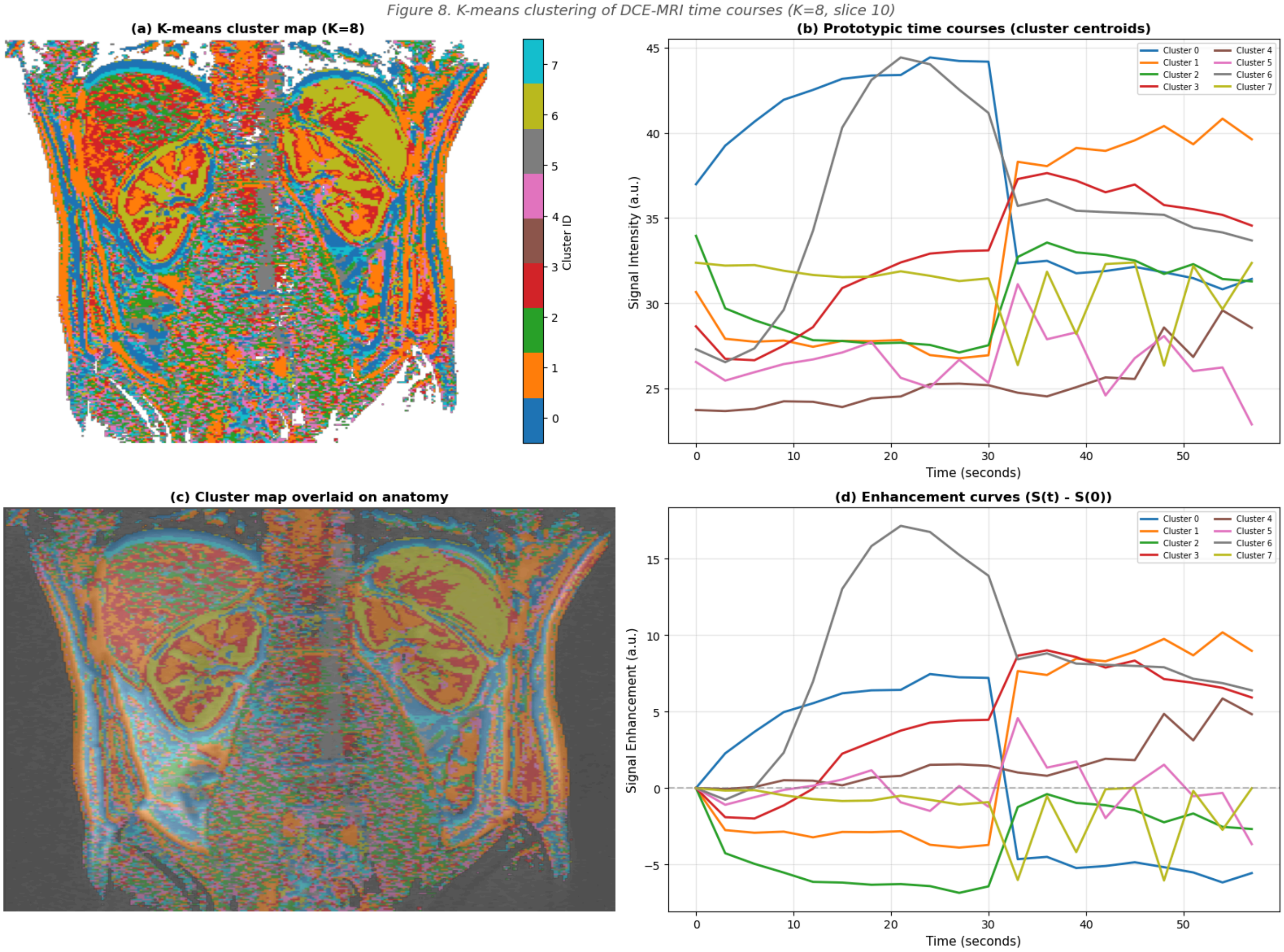
Given N data points (voxel time courses) $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\} \subset \mathbf{R}^{N_t}$ and a desired number of clusters K :

1. **Initialize** K cluster centroids $\{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K\}$ (e.g., K-means++ initialization)
2. **Assign** each point to its nearest centroid: $c_i = \arg\min_k \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$
3. **Update** centroids as the mean of assigned points: $\boldsymbol{\mu}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i$
4. **Repeat** steps 2–3 until convergence (assignments stop changing)

The centroids $\boldsymbol{\mu}_k$ represent **prototypic time courses** — one for each tissue type identified by the algorithm.

```
Running K-means with K=8 on 49,123 foreground voxels in R^20 feature space...
K-means converged in 62 iterations
  Cluster 0: 6,322 voxels (12.9%)
  Cluster 1: 9,608 voxels (19.6%)
  Cluster 2: 6,430 voxels (13.1%)
  Cluster 3: 7,071 voxels (14.4%)
  Cluster 4: 4,975 voxels (10.1%)
  Cluster 5: 5,355 voxels (10.9%)
  Cluster 6: 5,246 voxels (10.7%)
  Cluster 7: 4,116 voxels (8.4%)
```

Figure 8. K-means clustering of DCE-MRI time courses (K=8, slice 10)



9. K-means "Smoothing" of Time Courses

The noise problem

Individual voxel time courses in DCE-MRI are inherently **noisy** due to:

- Thermal noise in the MRI receiver coils
- Physiological noise (pulsation, motion residuals)
- Partial volume effects (voxels containing multiple tissue types)

K-means smoothing concept

K-means smoothing is an elegant approach to noise reduction:

1. Apply K-means clustering with a **large K** (e.g., K = 64) to the voxel time courses
2. Replace each voxel's noisy time course with its **cluster centroid** (the average of all voxels in that cluster)

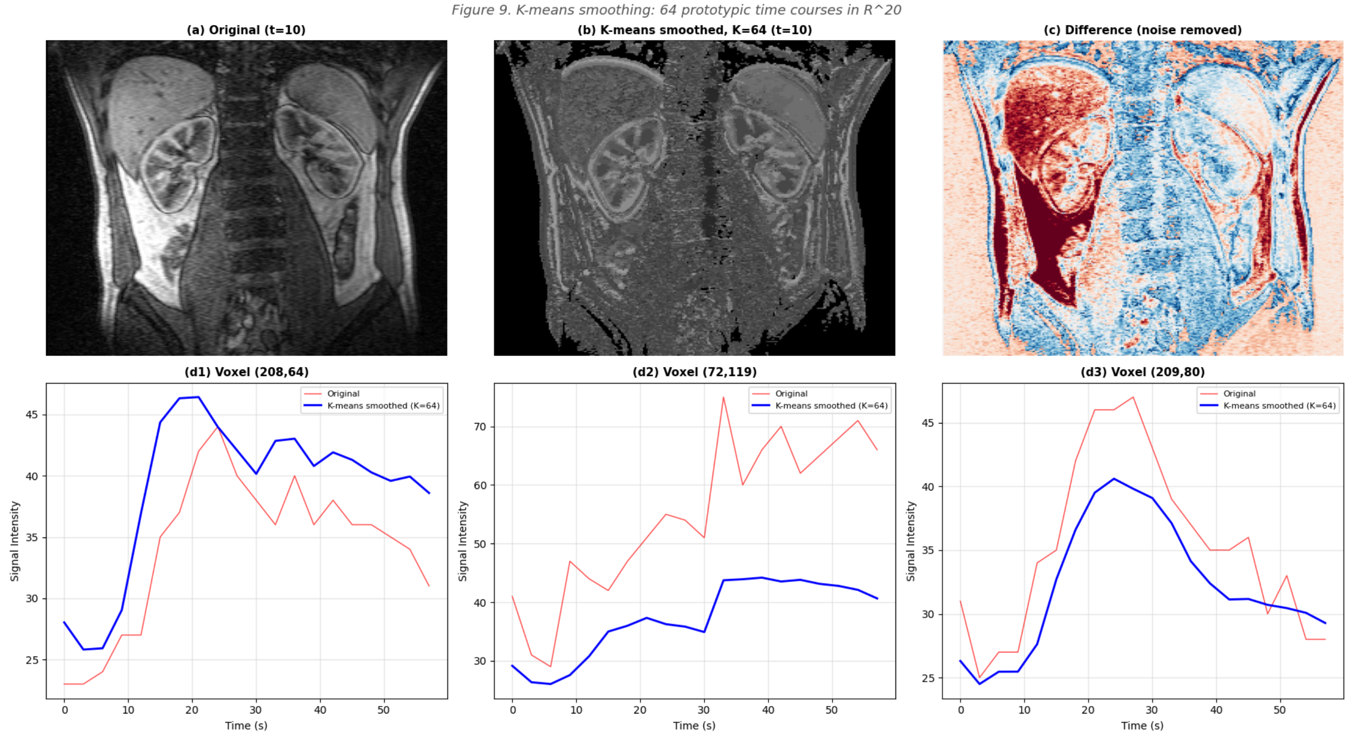
This is a form of **adaptive spatial smoothing** — it averages voxels that have similar temporal dynamics, respecting tissue boundaries. Unlike Gaussian spatial smoothing, which blurs across boundaries, K-means smoothing preserves the distinct enhancement patterns of cortex and medulla.

> **Key insight**: With K = 64 clusters and ~50 time points, each centroid is the average of many voxels (typically 50–500), reducing noise by a factor of ~$\sqrt{N_{cluster}}$ while preserving the tissue-specific signal dynamics.

The feature space is $\mathbf{R}^{N_t}$ (e.g., $\mathbf{R}^{50}$), and the 64 centroids represent **64 prototypic time courses** that capture the main patterns in the data.

```
Running K-means smoothing with K=64 clusters...
K-means converged in 181 iterations
  64 prototypic time courses in R^20
```

Figure 9. K-means smoothing: 64 prototypic time courses in R^20



DIY: K-means Clustering and Smoothing

Q6. In a DCE-MRI dataset with 50 time points, each voxel's time course is a vector in $\mathbf{R}^{50}$. If K-means with K=8 assigns 1,000 voxels to a cluster, what is the expected noise reduction factor for that cluster's centroid compared to a single voxel?

▶ Click to reveal answer

Q7. Why might you choose K=64 rather than, say, K=4 for K-means smoothing? What is the trade-off?

▶ Click to reveal answer

10. Tracer Kinetics and Compartment Modeling

Overview

The goal of compartment modeling in kidney DCE-MRI is to extract **physiological parameters** (perfusion, GFR, tubular transit time) from the observed signal intensity time courses. This requires:

1. Converting signal intensity $S(t)$ to contrast agent concentration $C(t)$
2. Measuring or estimating the **Arterial Input Function** $C_a(t)$
3. Fitting a **compartment model** to the tissue concentration curve

The General Tracer Kinetic Equation

For a linear, time-invariant system, the tissue concentration is related to the arterial input by a **convolution**:

$$C_{tissue}(t) = F_p \cdot C_a(t) \otimes R(t) = F_p \int_0^t C_a(\tau) \cdot R(t-\tau)\, d\tau$$

where:

- F_p = plasma flow (mL/min/mL tissue)
- $C_a(t)$ = arterial input function
- $R(t)$ = **impulse residue function** (the fraction of tracer still in the tissue at time t after an instantaneous input)
- $\otimes$ = convolution operator

Different compartment models correspond to different forms of $R(t)$.

The Two-Compartment Filtration Model for Kidney

For the kidney, a commonly used model has two tissue compartments:

1. **Plasma compartment** (volume fraction v_p): Blood plasma in glomerular and peritubular capillaries
2. **Tubular compartment** (volume fraction v_t): Filtrate in the tubules

The model equations are:

$$\frac{d\, v_p \cdot C_p(t)}{dt} = F_p \cdot [C_a(t) - C_p(t)] - GFR \cdot C_p(t)$$

$$\frac{d\, v_t \cdot C_t(t)}{dt} = GFR \cdot C_p(t) - T_t^{-1} \cdot v_t \cdot C_t(t)$$

$$C_{tissue}(t) = v_p \cdot C_p(t) + v_t \cdot C_t(t)$$

where:

- F_p = plasma flow rate per unit tissue volume (mL/min/mL)

- GFR = glomerular filtration rate per unit tissue volume (mL/min/mL)
- T_t = mean tubular transit time (min)

The **Filtration Fraction** is $FF = GFR/F_p$.

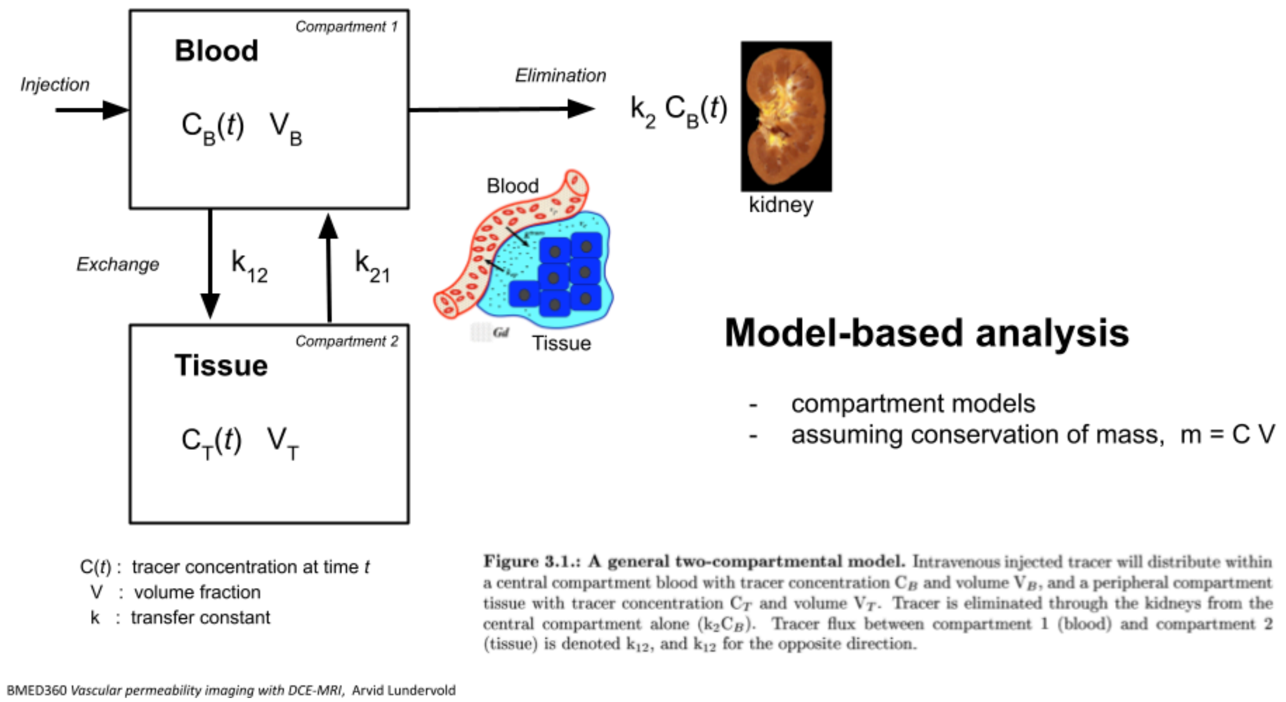


Figure. Model-based analysis of DCE-MRI data. The concentration-time curve in each voxel (or ROI) is fitted to a pharmacokinetic compartment model, yielding quantitative parameter maps (e.g., perfusion, GFR, transit time). The arterial input function (AIF) serves as the driving input to the model.

Source: BMED360 Lecture 6 — Vascular permeability and DCE-MRI (A. Lundervold, UiB).

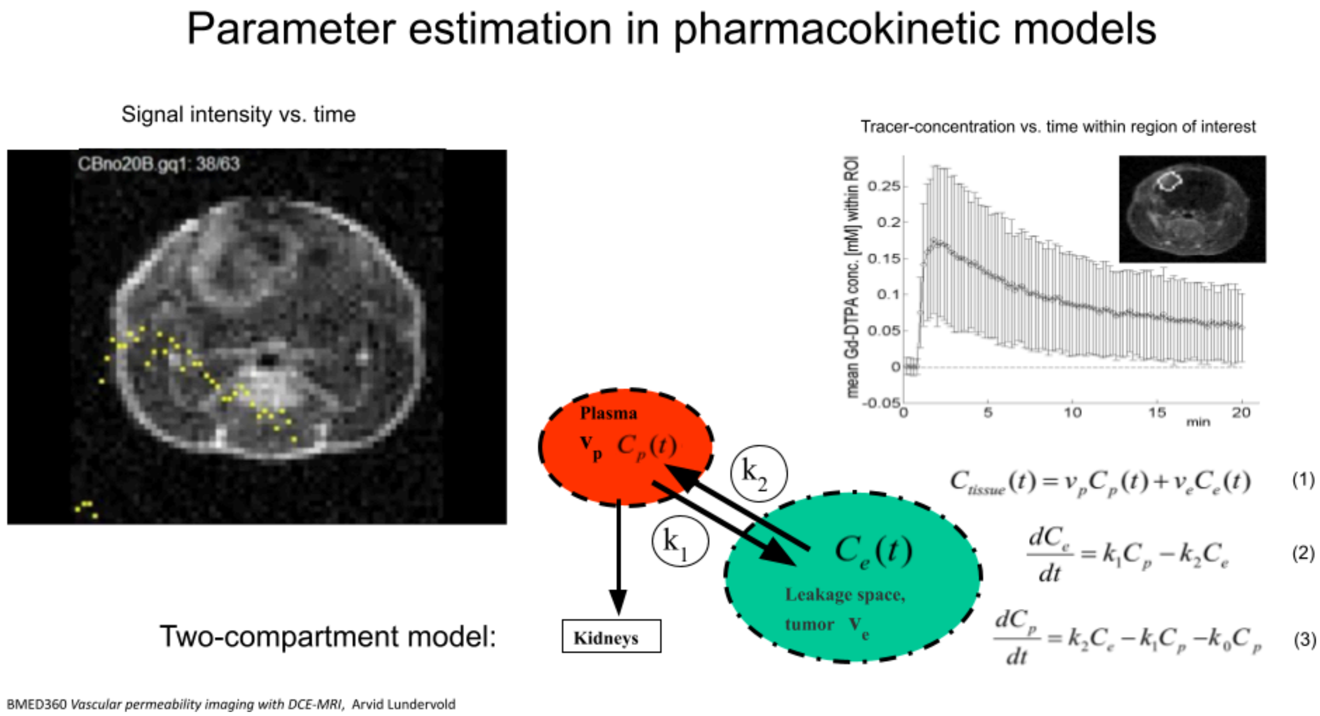
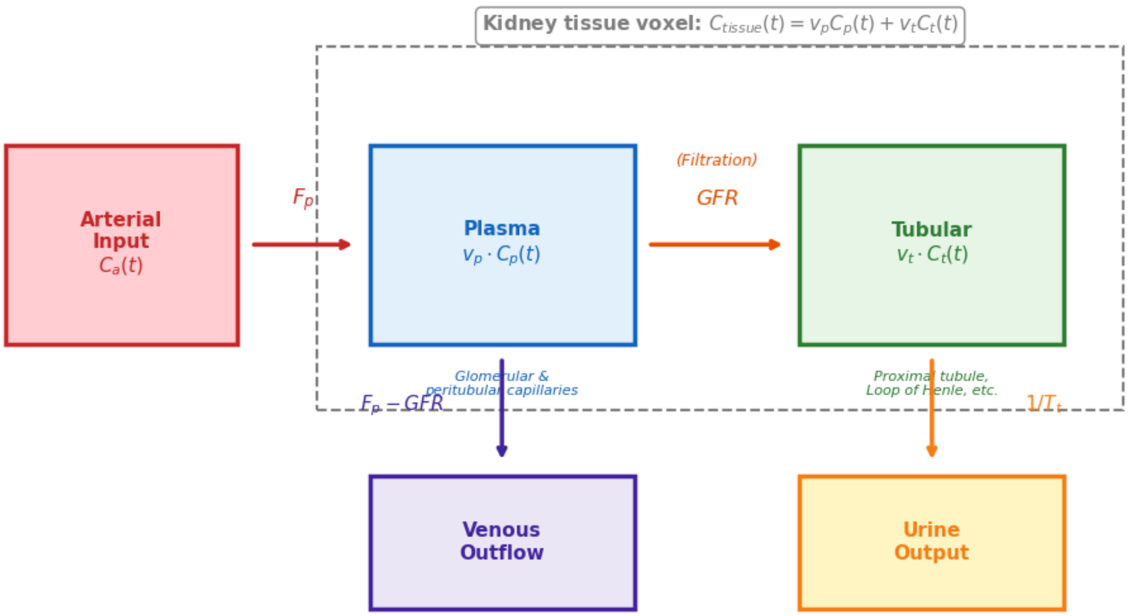


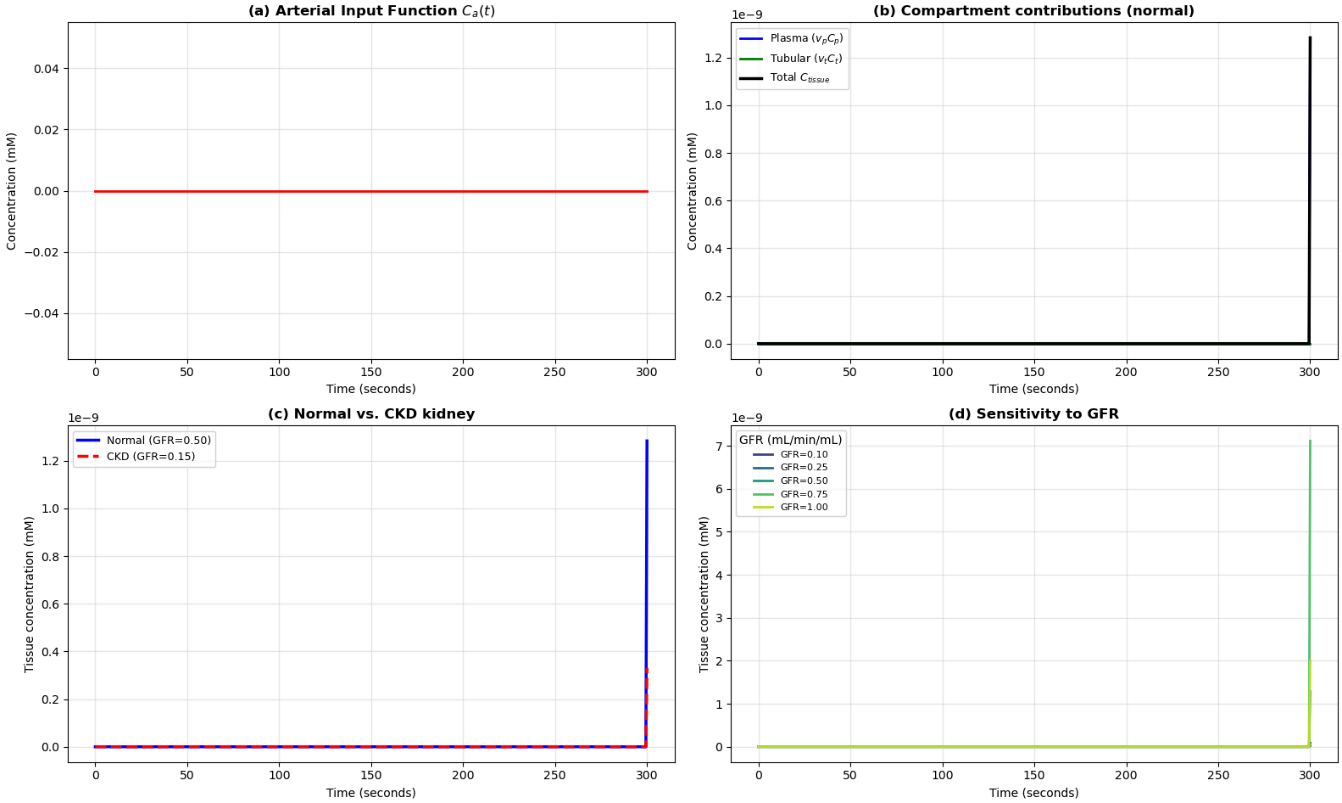
Figure. Pharmacokinetic compartment model for DCE-MRI analysis. The model describes the exchange of contrast agent between the vascular (plasma) space and the extravascular extracellular space (EES), governed by the transfer constant K^{trans} and the rate constant k_{ep}. For the kidney, the "EES" is replaced by the **tubular compartment**, and K^{trans} corresponds to the **GFR** (per unit tissue volume).
Source: BMED360 Lecture 6 — Vascular permeability and DCE-MRI (A. Lundervold, UiB).

Figure 10. Two-compartment filtration model for kidney DCE-MRI




```
Normal kidney cortex parameters:
  Plasma flow (Fp):        3.0 mL/min/mL
  GFR density:             0.50 mL/min/mL
  Filtration fraction (FF): 16.7%
  Plasma volume (vp):      0.15
  Tubular volume (vt):     0.10
  Tubular transit time:    2.0 min
```

Figure 11. Two-compartment kidney model simulation




```
============================================================
Compartment Model Fitting Results
============================================================
Parameter         True  Estimated  Std Error   Error%
------------------------------------------------------------
  Fp             3.000      1.997      0.000    33.4%
  GFR_d          0.500      0.300      0.000    40.0%
  vp             0.150      0.113      0.248    24.4%
  vt             0.100      0.100      0.000     0.0%
  Tt             2.000      1.500      0.000    25.0%

  Single-kidney GFR (V_kidney = 150 mL):
    True:      75.0 mL/min
    Estimated: 45.0 mL/min
```

Figure 12. GFR estimation from compartment model fitting

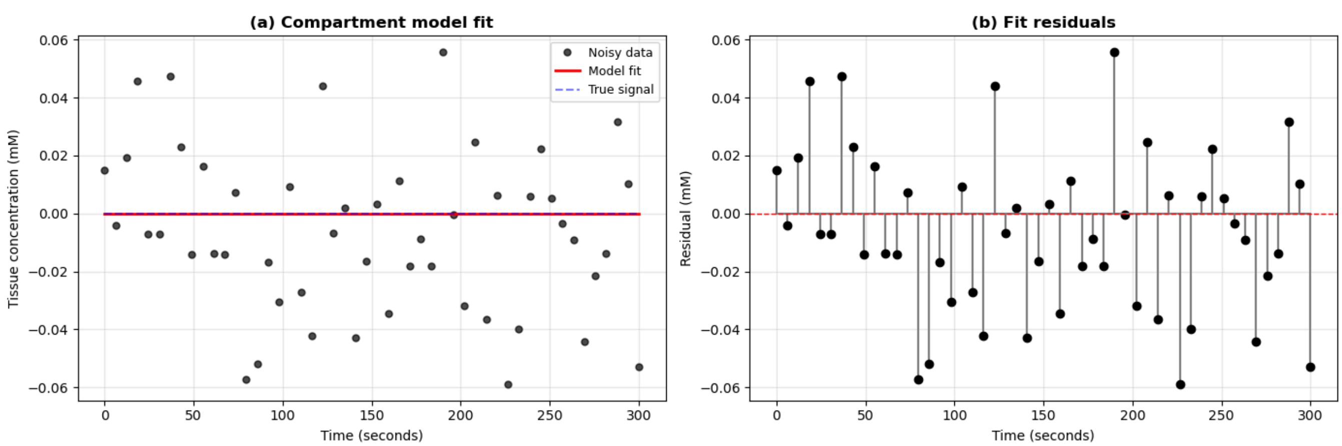
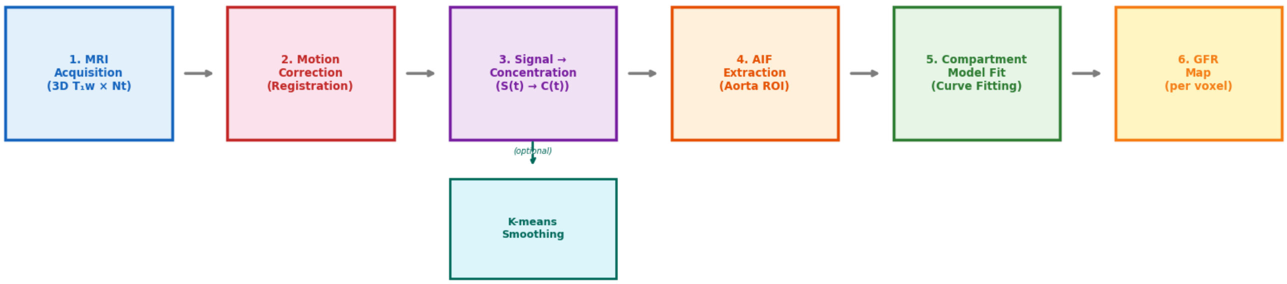


Figure 13. Complete DCE-MRI pipeline for kidney GFR estimation



DIY: Tracer Kinetics and Compartment Modeling

Q8. In the two-compartment kidney model, what happens to the tissue concentration curve $C_{tissue}(t)$ if GFR is very small (approaching zero)?

▶ Click to reveal answer

Q9. What is the **Arterial Input Function (AIF)** and why is it critical for quantitative DCE-MRI?

▶ Click to reveal answer

Q10. If a healthy kidney has a volume of 150 mL, a plasma flow of 3.0 mL/min/mL, and a filtration fraction of 20%, what is the estimated single-kidney GFR?

▶ Click to reveal answer

11. Putting It All Together: From Voxels to GFR Maps

In a full clinical DCE-MRI analysis, the steps would be:

1. **Motion correction** — register all time points to a reference volume
2. **Kidney segmentation** — identify cortex and medulla regions
3. **K-means smoothing** (optional) — reduce noise with K=64 prototypic time courses
4. **AIF extraction** — measure $C_a(t)$ from the aorta
5. **Voxel-wise or ROI-based model fitting** — fit the two-compartment model to each voxel/ROI
6. **Parameter maps** — display GFR, perfusion, and transit time as color maps overlaid on anatomy

Below, we demonstrate a simplified version: fitting the compartment model to the K-means cluster centroids to estimate mean GFR for each tissue cluster.

```
Fitting two-compartment model to K=8 cluster centroids...

 Cluster       Fp    GFR_d       vp       vt       Tt     skGFR*
-----------------------------------------------------------------
      0     2.00    0.300    0.100    0.050     1.50      24.0
      1     2.00    0.300    0.100    0.050     1.50      24.0
      2     2.00    0.300    0.100    0.050     1.50      24.0
      3     2.00    0.300    0.100    0.050     1.50      24.0
      4     2.00    0.300    0.100    0.050     1.50      24.0
      5     2.00    0.300    0.100    0.050     1.50      24.0
      6     2.00    0.300    0.100    0.050     1.50      24.0
      7     2.00    0.300    0.100    0.050     1.50      24.0

* skGFR = GFR_density x 80 mL (assumed cortical volume)
  Note: These are illustrative values; accurate GFR requires proper
  signal-to-concentration conversion and AIF measurement.
```

12. Summary and Going Deeper

What we covered

This notebook explored the full pipeline of **DCE-MRI-based kidney GFR estimation** as a case study in computational medicine:

Topic	Key concepts
DCE-MRI basics	Gd contrast injection, T\u2081 enhancement, 4D data, signal-to-concentration conversion
Definitions	GFR, AIF, compartment models, tracer kinetics, impulse residue function
Historical context	From indicator dilution theory (1927) to modern AI-assisted analysis (2020s)
Data exploration	Loading NIfTI files, temporal montage, ROI-based time courses
Motion correction	Respiratory motion, rigid/deformable registration, center-of-mass tracking
K-means clustering	Voxel time courses as feature vectors in $\mathbf{R}^{N_t}$, tissue segmentation
K-means smoothing	64 prototypic time courses, adaptive noise reduction
Compartment modeling	Two-compartment (plasma + tubular) ODE model, GFR as a model parameter
GFR estimation	Curve fitting, parameter maps, single-kidney GFR

Extensions and advanced topics

1. **Proper AIF extraction**: Automated detection of the aorta, partial volume correction, T2* correction at high concentrations
2. **Non-linear signal-to-concentration conversion**: Using the full SPGR signal equation with pre-contrast T\u2081 mapping

3. **Bayesian inference**: Incorporating prior physiological knowledge into parameter estimation to improve robustness
4. **Deep learning approaches**: Neural networks for automated kidney segmentation, motion correction, and parameter estimation
5. **Multi-site standardization**: The PARENCHIMA COST Action and UKRIN-MAPS project for standardizing renal MRI protocols
6. **Radiomics and texture analysis**: Extracting higher-order features from DCE-MRI parameter maps for phenotyping kidney disease
7. **Diffusion-weighted MRI (DWI)**: Complementary technique for assessing renal microstructure without contrast agents

Key references

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3. **Hodneland, E., Hanson, E.A., Lundervold, A., Modersitzki, J., Eikefjord, E. & Munthe-Kaas, A.Z.** (2014). Segmentation-driven image registration — application to 4D DCE-MRI recordings of the moving kidneys. *IEEE Trans. Image Process.* 23:2392–2404.
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7. **Layton, A.T. & Edwards, A.** (2014). *Mathematical Modeling in Renal Physiology*. Springer.

Software tools

- **MRTOOL** / **PMI** — MATLAB-based tools for DCE-MRI analysis (Sourbron group)
- **DCE@urLAB** — Open-source Python/MATLAB package for DCE-MRI pharmacokinetic modeling
- **ROCKETSHIP** — ImageJ/MATLAB plugin for DCE-MRI analysis
- **nibabel** + **scipy** + **scikit-learn** — The Python ecosystem used in this notebook

13. DIY Exercises: Explore Further

Exercise 1: Vary K in K-means Clustering

Run the K-means clustering with different values of K (e.g., K = 4, 16, 32, 64, 128). How does the cluster map change? How does it affect the smoothed time courses?

▶ Click for starter code

▶ Click for what you should observe

Exercise 2: Explore Different Slices

The kidney anatomy varies dramatically across slices. Apply the analysis to different slices (not just the middle one) and compare the results.

▶ Click for starter code

Exercise 3: Sensitivity Analysis of Compartment Model

Explore how the compartment model output changes when you vary individual parameters. This helps build intuition for what the model "sees" in the data.

▶ Click for starter code

Notebook summary: This notebook explored **DCE-MRI of the kidney** as a computational medicine approach to non-invasive GFR estimation. We covered the physics of DCE-MRI acquisition, the mathematics of tracer kinetic compartment models, practical aspects of motion correction and K-means clustering/smoothing, and the clinical significance of image-based kidney function assessment. The notebook builds on 06-kidney-filtration.ipynb (physiology and mathematical modeling of GFR) by adding the imaging dimension — turning abstract filtration rates into spatially-resolved, patient-specific functional maps.

Created for BMED365 — Computational Medicine, University of Bergen, 2026.

See also:

- functional-kidney-imaging — ESMRMB 2016 teaching materials on renal perfusion and filtration with MRI
- BMED360-2021 Lab5-DCE-MRI — Original source notebook
- PARENCHIMA — European network for standardization of renal MRI